

IF THE GAUGE MOVES – THIS IS THE NEW BIT

Existing appliances connected • Natural Gas

FOLLOW THESE STEPS

1. GAUGE MOVES?

Any perceptible fall during the test?



YES

If the drop is more than 1 mbar you must calculate the **Installation Volume (IV)**.

CALCULATE IV

Use the IV calculator or tables.

PIPE LENGTHS
15 / 22 / 28 mm
(no 35 mm)



METER IV
G4 / U6
or E6

=

IV
(m³)



Next slide shows how to calculate IV (step by step).

2. CHECK THE TABLE

Find the maximum permissible drop for your IV.

Installation Volume (IV)	Max drop (mbar)
≤ 0.005 m ³	8
> 0.005 – ≤ 0.010 m ³	4
> 0.010 – ≤ 0.015 m ³	2.5
> 0.015 – ≤ 0.035 m ³	1

Example:

- IV = 0.014 m³
- This falls in > 0.010 – ≤ 0.015 m³
- Maximum permissible drop = **2.5 mbar**

3. APPLY YOUR RESULT

Compare the drop on the gauge with the table.



Observed drop:

2 mbar

In this example:

- IV = 0.014 m³
- Maximum permitted drop = 2.5 mbar
- Observed drop (2 mbar) is **LESS** than 2.5 mbar



WITHIN PERMISSIBLE LIMIT
OK so far

4. NEXT STEP – TEST PIPEWORK

Isolate all appliances and repeat the test on the pipework only.



Repeat the tightness test with all appliances isolated.



ZERO MOVEMENT
= **PASS**



ANY MOVEMENT
= **FAIL**

Trace, repair and make safe.
Re-test until zero movement.



PURGING AFTER WORK?

PV = 1.5 × IV

0.014336 × 1.5 = 0.021504 m
Purge at least 0.022 m

FINISH: Reconnect appliances → gas ON → LDF test all disturbed joints + pipework between isolation valve and appliance.

STEP 1 – SURVEY THE PIPEWORK

Walk the property and break the gas pipework into simple sections.

MEASURE IT RIGHT

ALL PIPEWORK COUNTS

1. IDENTIFY THE METER TYPE

Use the table below to get the meter IV.

Meter type	Meter IV (m ³)
G4 / U6	0.008
E6	0.0024

G4 / U6



0.008 m³

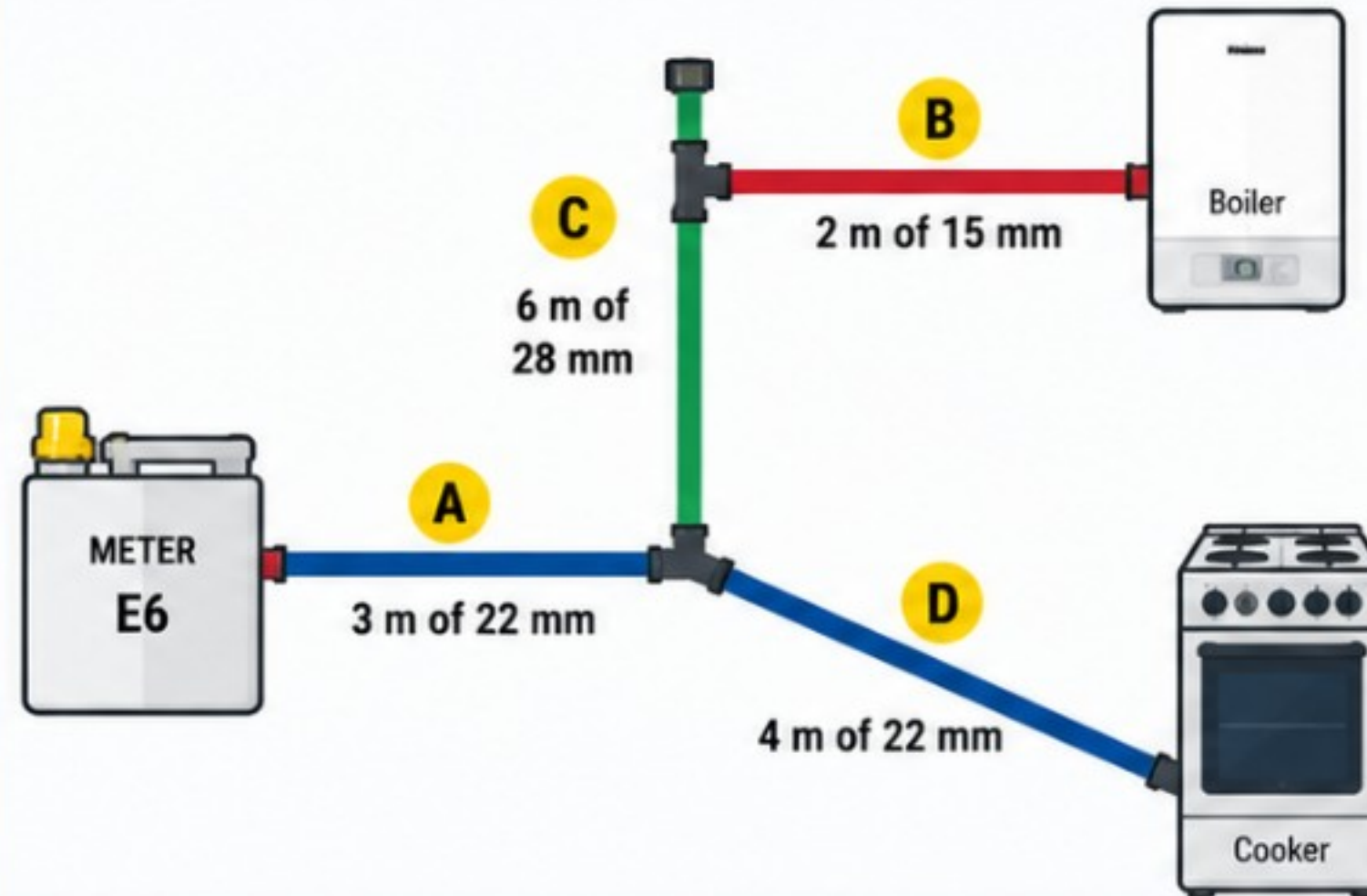
E6



0.0024 m

2. EXAMPLE PIPEWORK LAYOUT

Break the pipework into simple sections and measure the length of each.



WHAT TO WRITE DOWN

- 1 Meter type (G4/U6 or E6)
- 2 Pipe size (e.g. 15 mm, 22 mm, 28 mm)
- 3 Length of each pipe section (in metres)
- 4 Any inaccessible / unknown areas
- 5 Appliances to isolate later

i REMEMBER

- Measure ALL pipework from the meter to each appliance.
- Include horizontal and vertical runs.
- Split by pipe size (e.g. all 15 mm together, all 22 mm together).
- Use reasonable estimates if not accessible and make a note.

1 SPLIT BY SIZE



Add all 15 mm together, all 22 mm together, etc.

2 MEASURE LENGTH



Measure the pipe length in metres. Horizontal and vertical runs both count.

3 PUT IT INTO THE CALCULATOR



Enter your lengths and meter type into the IV calculator. It will do the maths and give you the total IV.

STEP 2 – HOW THE IV IS WORKED OUT

Add the meter volume, pipework volume and 10% for fittings = Total Installation Volume (IV)

SIMPLE SUM
USE THE TABLES / CALCULATOR

1. METER IV

Use the meter table below (common sizes).

Meter type	Meter IV (m³)
G4 / U6	0.008
E6	0.0024

G4 / U6



0.008 m³

E6



0.0024 m



Check the meter label if unsure.

Use the IV value from the table or the IV calculator.

2. PIPEWORK IV

Measure each length of pipe and multiply by the volume per metre.

Pipe size	Volume per metre (m³)	Example length	Example IV
15 mm	0.00014	2 m	0.00028
22 mm	0.00032	7 m	0.00224
28 mm	0.00054	6 m	0.00324

Pipework total = 0.00576 m³

3. ADD 10% FOR FITTINGS

Add 10% to the pipework total to allow for fittings, bends, tees etc.

10% of 0.00576 m = 0.000576 m

4. TOTAL INSTALLATION VOLUME (IV)

Add together the meter IV, pipework IV and 10% for fittings.

Meter IV (e.g. G4 / U6 or E6)	0.00800 m³
Pipework IV total	0.00576 m³
10% for fittings	0.000576 m
TOTAL IV	0.014336 m

SIMPLE STEP-BY-STEP

- 1 Find your meter IV (look at the meter label).
- 2 Measure the pipe lengths in the property.
- 3 Multiply each length by the volume per metre.
- 4 Add the pipework volumes together.
- 5 Add 10% for fittings.
- 6 Add the meter IV.
- 7 This gives you the Total Installation Volume (IV).

EXAMPLE RESULT



Total IV = 0.014336 m
(G4 / U6 meter)

USE THE IV



Now compare this IV with the permissible drop table (Step 3).

STEP 3 – USE THE IV TO DECIDE PASS OR FAIL

Compare your Installation Volume (IV) with the table and check the pressure drop.

NATURAL GAS

EXAMPLE SHOWN

1. NATURAL GAS PERMISSIBLE DROP TABLE

Installation Volume (IV)	Maximum pressure drop
$\leq 0.005 \text{ m}^3$	8 mbar
$> 0.005 - \leq 0.010 \text{ m}^3$	4 mbar
$> 0.010 - \leq 0.015 \text{ m}^3$	2.5 mbar
$> 0.015 - \leq 0.035 \text{ m}^3$	1 mbar

2. APPLY YOUR RESULT

Example: IV = 0.014 m^3



Observed drop
on gauge:

2 mbar

Compare with table:

- IV 0.014 m^3 falls in the $> 0.010 - \leq 0.015 \text{ m}^3$ band
- Maximum permitted drop = **2.5 mbar**
- Observed drop (2 mbar) is LESS than 2.5 mbar → **OK so far**

3. NEXT STEP

Isolate all appliances and repeat the test on the pipework.



Repeat tightness test
on pipework only



PIPEWORK TEST RESULT

**ZERO MOVEMENT
= PASS**

No perceptible movement on the gauge
with appliances isolated.



PIPEWORK TEST RESULT

**ANY MOVEMENT
= FAIL**

Trace, repair and make safe.
Re-test until zero movement.



PURGING AFTER WORK?

PV = $1.5 \times \text{IV}$

$0.014336 \times 1.5 = 0.021504 \text{ m}^3$

Purge at least 0.022 m^3